Machine Learning
CMPSCI 589

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Naive Bayes (2/2)
Bernoulli Naive Bayes

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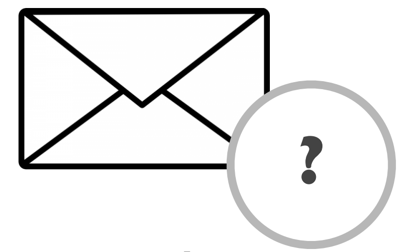
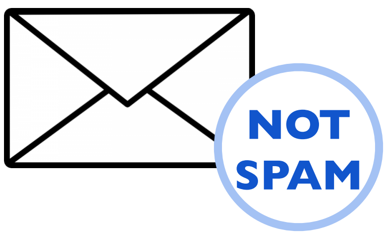
Naive Bayes Algorithm

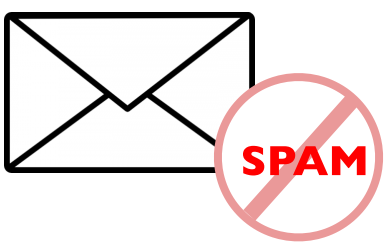
Review: Most Likely Class/Label

- Intuition

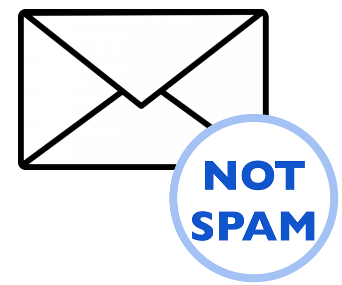
New email







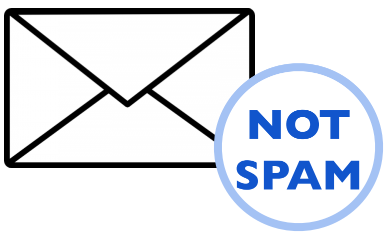
e-mail	Contents	Class
1	"password expired"	SPAM
2	"send password"	SPAM
3	"review conference"	NOT SPAM
4	"password review"	SPAM
5	"review account"	SPAM
6	"account password"	SPAM
7	"send account"	SPAM
8	"conference paper"	NOT SPAM
9	"send paper"	NOT SPAM
10	"expired account"	SPAM

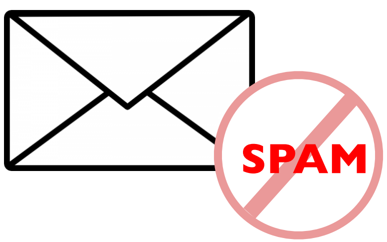


"Given a new email whose contents are "*review conference paper*",
it is more likely to be a SPAM or NOT SPAM?

Review: Most Likely Class/Label

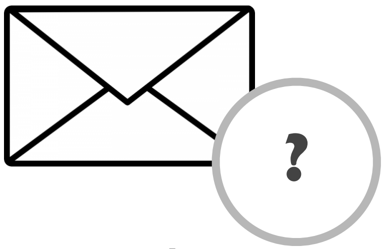
- Intuition

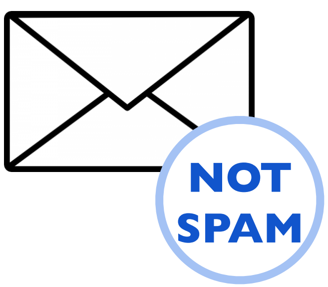




e-mail	Contents	Class
1	"password expired"	SPAM
2	"send password"	SPAM
3	"review conference"	NOT SPAM
4	"password review"	SPAM
5	"review account"	SPAM
6	"account password"	SPAM
7	"send account"	SPAM
8	"conference paper"	NOT SPAM
9	"send paper"	NOT SPAM
10	"expired account"	SPAM

New email





What is the most likely class of the email?

Review: Naive Bayes Classifier

$$\Pr(y_i \mid \mathbf{x}) = \frac{\Pr(y_i)\ \Pr(x_1, \ldots, x_n \mid y_i)}{\Pr(x_1, \ldots, x_n)}$$

How to compute these probabilities based on a training set?

y_1 = spam
y_2 = not_spam
$\mathbf{x} = (x_1, \ldots x_n)$ = words in email

Review: Naive Bayes Classifier

y_1 = **spam**
y_2 = **not_spam**
$\mathbf{x} = (x_1, \ldots x_n)$ = **words in email**

$$\Pr(y_i \mid \mathbf{x}) = \frac{\Pr(y_i)\ \Pr(x_1, \ldots, x_n \mid y_i)}{\Pr(x_1, \ldots, x_n)}$$

$$\left.\begin{array}{l} \Pr(y = \textbf{spam} \mid x_1 = \textbf{inheritance}, x_2 = \textbf{millions}) \\[2em] \Pr(y = \textbf{not\_spam} \mid x_1 = \textbf{inheritance}, x_2 = \textbf{millions}) \end{array}\right\} \text{max}$$

in practice

Review: Naive Bayes Classifier

y_1 = spam
y_2 = not_spam
$\mathbf{x} = (x_1, \ldots x_n)$ = words in email

$$\Pr(y_i \mid \mathbf{x}) = \frac{\Pr(y_i)\ \Pr(x_1, \ldots, x_n \mid y_i)}{\Pr(x_1, \ldots, x_n)}$$

$$\Pr(y = \textbf{spam} \mid x_1 = \textbf{inheritance}, x_2 = \textbf{millions}) \propto \Pr(y = \textbf{spam}) \times$$
$$\Pr(x_1 = \textbf{inheritance} \mid y = \textbf{spam})\ \Pr(x_2 = \textbf{millions} \mid y = \textbf{spam})$$

$$\Pr(y = \textbf{not\_spam} \mid x_1 = \textbf{inheritance}, x_2 = \textbf{millions}) \propto \Pr(y = \textbf{not\_spam}) \times$$
$$\Pr(x_1 = \textbf{inheritance} \mid y = \textbf{not\_spam})\ \Pr(x_2 = \textbf{millions} \mid y = \textbf{not\_spam})$$

Review: Naive Bayes Classifier

y_1 = **spam**
y_2 = **not_spam**
$\mathbf{x} = (x_1, \ldots x_n)$ = **words in email**

$$\Pr(y_i \,|\, \mathbf{x}) = \frac{\Pr(y_i)\ \Pr(x_1,\ \ldots, x_n \mid y_i)}{\Pr(x_1,\ \ldots, x_n)}$$

$$\Pr(y = \textbf{spam} \,|\, x_1 = \textbf{inheritance}, x_2 = \textbf{millions}) \propto \boxed{\Pr(y = \textbf{spam})} \times \Pr(x_1 = \textbf{inheritance} \,|\, y = \textbf{spam})\ \Pr(x_2 = \textbf{millions} \,|\, y = \textbf{spam})$$

$$\boxed{\frac{\textbf{spams}}{\textbf{emails}}} \times \frac{\textbf{spams containing inheritance}}{\textbf{spams}} \times \frac{\textbf{spams containing millions}}{\textbf{spams}}$$

Review: Naive Bayes Classifier

y_1 = **spam**
y_2 = **not_spam**
$\mathbf{x} = (x_1, \ \ldots \ x_n)$ = **words in email**

$$\Pr(y_i \mid \mathbf{x}) = \frac{\Pr(y_i) \ \Pr(x_1, \ \ldots, x_n \mid y_i)}{\Pr(x_1, \ \ldots, x_n)}$$

$$\Pr(y = \textbf{spam} \mid x_1 = \textbf{inheritance}, x_2 = \textbf{millions}) \ \propto \ \Pr(y = \textbf{spam}) \times \boxed{\Pr(x_1 = \textbf{inheritance} \mid y = \textbf{spam})} \Pr(x_2 = \textbf{millions} \mid y = \textbf{spam})$$

$$\frac{\textbf{spams}}{\textbf{emails}} \times \boxed{\frac{\textbf{spams containing inheritance}}{\textbf{spams}}} \times \frac{\textbf{spams containing millions}}{\textbf{spams}}$$

Review: Naive Bayes Classifier

y_1 = **spam**
y_2 = **not_spam**
$\mathbf{x} = (x_1, \ldots x_n)$ = **words in email**

$$\Pr(y_i \mid \mathbf{x}) = \frac{\Pr(y_i)\ \Pr(x_1, \ldots, x_n \mid y_i)}{\Pr(x_1, \ldots, x_n)}$$

$$\Pr(y = \textbf{spam} \mid x_1 = \textbf{inheritance}, x_2 = \textbf{millions}) \propto \Pr(y = \textbf{spam}) \times \Pr(x_1 = \textbf{inheritance} \mid y = \textbf{spam}) \boxed{\Pr(x_2 = \textbf{millions} \mid y = \textbf{spam})}$$

$$\frac{\textbf{spams}}{\textbf{emails}} \times \frac{\textbf{spams containing inheritance}}{\textbf{spams}} \times \boxed{\frac{\textbf{spams containing millions}}{\textbf{spams}}}$$

Review: Naive Bayes Classifier

y_1 = spam
y_2 = not_spam
$\mathbf{x} = (x_1, \ldots x_n)$ = words in email

$$\Pr(y_i \,|\, \mathbf{x}) = \frac{\Pr(y_i)\ \Pr(x_1, \ldots, x_n \mid y_i)}{\Pr(x_1, \ldots, x_n)}$$

$$\Pr(y = \textbf{spam} \,|\, x_1 = \textbf{inheritance}, x_2 = \textbf{millions}) \ \propto$$

$$\frac{\textbf{spams}}{\textbf{emails}} \times \frac{\textbf{spams containing inheritance}}{\textbf{spams}} \times \frac{\textbf{spams containing millions}}{\textbf{spams}}$$

$$\Pr(y = \textbf{not\_spam} \,|\, x_1 = \textbf{inheritance}, x_2 = \textbf{millions}) \ \propto$$

$$\frac{\textbf{not\_spams}}{\textbf{emails}} \times \frac{\textbf{not\_spams containing inheritance}}{\textbf{not\_spams}} \times \frac{\textbf{not\_spams containing millions}}{\textbf{not\_spams}}$$

in practice

Naive Bayes Classifier - Example

$$y := \underset{y_i \in Y}{\arg\max} \; \Pr(y_i) \prod_{k=1}^{n} \Pr(x_k \mid y_i)$$

Instance #	*Color*	*Type*	*Made_In*	**Class/Label: Likely to be stolen?**
1	Red	Convertible	USA	**Yes**
2	Red	Convertible	USA	**No**
3	Red	Convertible	USA	**Yes**
4	Yellow	Convertible	USA	**No**
5	Yellow	Convertible	Imported	**Yes**
6	Yellow	SUV	Imported	**No**
7	Yellow	SUV	Imported	**Yes**
8	Yellow	SUV	USA	**No**
9	Red	Convertible	Imported	**No**
10	Red	Convertible	Imported	**Yes**

- We wish to predict: will a red SUV made in the U.S. be stolen?

Naive Bayes Classifier - Example

$$y := \underset{y_i \in Y}{\arg\max} \; \Pr(y_i) \prod_{k=1}^{n} \Pr(x_k \mid y_i)$$

Instance #	*Color*	*Type*	*Made_In*	**Class/Label: Likely to be stolen?**
1	Red	Convertible	USA	**Yes**
2	Red	Convertible	USA	**No**
3	Red	Convertible	USA	**Yes**
4	Yellow	Convertible	USA	**No**
5	Yellow	Convertible	Imported	**Yes**
6	Yellow	SUV	Imported	**No**
7	Yellow	SUV	Imported	**Yes**
8	Yellow	SUV	USA	**No**
9	Red	Convertible	Imported	**No**
10	Red	Convertible	Imported	**Yes**

will a
red SUV made in the U.S.
be stolen?

Pr(Stolen=yes | color=Red, type=SUV, made_in=USA)

Pr(Stolen=no | color=Red, type=SUV, made_in=USA)

Naive Bayes Classifier - Example

Instance #	*Color*	*Type*	*Made_In*	Class/Label: Likely to be stolen?
1	Red	Convertible	USA	Yes
2	Red	Convertible	USA	No
3	Red	Convertible	USA	Yes
4	Yellow	Convertible	USA	No
5	Yellow	Convertible	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	USA	No
9	Red	Convertible	Imported	No
10	Red	Convertible	Imported	Yes

will a
red SUV made in the U.S.
get stolen?

$$\Pr(\text{Stolen=yes} \mid \text{color=Red, type=SUV, made\_in=USA}) = \frac{5}{10} \times \frac{3}{5} \times \frac{2}{5} \times \frac{1}{5} = \frac{30}{10 \times 5^3}$$

$$= \Pr(\text{Stolen}) \times \Pr(\text{Red} \mid \text{Stolen}) \times \Pr(\text{USA} \mid \text{Stolen}) \times \Pr(\text{SUV} \mid \text{Stolen})$$

$$\Pr(\text{Stolen=Yes}) = \frac{5}{10} \qquad \Pr(\text{Made\_In = USA} \mid \text{Stolen=Yes}) = \frac{2}{5}$$

$$\Pr(\text{Color=Red} \mid \text{Stolen=Yes}) = \frac{3}{5} \qquad \Pr(\text{Type = SUV} \mid \text{Stolen=Yes}) = \frac{1}{5}$$

$$\Pr(\text{Stolen=no} \mid \text{color=Red, type=SUV, made\_in=USA}) = \frac{5}{10} \times \frac{2}{5} \times \frac{3}{5} \times \frac{2}{5} = \frac{60}{10 \times 5^3}$$

$$= \Pr(\text{Not\_Stolen}) \times \Pr(\text{Red} \mid \text{Not\_Stolen}) \times \Pr(\text{USA} \mid \text{Not\_Stolen}) \times \Pr(\text{SUV} \mid \text{Not\_Stolen})$$

$$\Pr(\text{Stolen=No}) = \frac{5}{10} \qquad \Pr(\text{Made\_In = USA} \mid \text{Stolen=No}) = \frac{3}{5}$$

$$\Pr(\text{Color=Red} \mid \text{Stolen=No}) = \frac{2}{5} \qquad \Pr(\text{Type = SUV} \mid \text{Stolen=No}) = \frac{2}{5}$$

Naive Bayes Classifier - Problem #1

Instance #	*Color*	*Type*	*Made_In*	**Class/Label: Likely to be stolen?**
1	Red	Convertible	USA	**Yes**
2	Red	Convertible	USA	**No**
3	Red	Convertible	USA	**Yes**
4	Yellow	Convertible	USA	**No**
5	Yellow	Convertible	Imported	**Yes**
6	Yellow	SUV	Imported	**No**
7	Yellow	SUV	Imported	**Yes**
8	Yellow	SUV	USA	**No**
9	Red	Convertible	Imported	**No**
10	Red	Convertible	Imported	**Yes**

will a
red SUV made in the U.S.
get stolen?

$$\Pr(\text{Stolen=yes} \mid \text{color=Red, type=SUV, made\_in=USA})$$

$$= \Pr(\text{Stolen}) \times \Pr(\text{Red} \mid \text{Stolen}) \times \Pr(\text{USA} \mid \text{Stolen}) \times \Pr(\text{SUV} \mid \text{Stolen})$$

$$\Pr(\text{Stolen=Yes}) = \frac{5}{10}$$

$$\Pr(\text{Made\_In = USA} \mid \text{Stolen=Yes}) = \frac{2}{5}$$

$$\Pr(\text{Color=Red} \mid \text{Stolen=Yes}) = ?$$

$$\Pr(\text{Type = SUV} \mid \text{Stolen=Yes}) = \frac{1}{5}$$

What would happen if there were no Red examples in the training set?

Naive Bayes Classifier - Problem #1

Instance #	Color	Type	Made_In	Class/Label: Likely to be stolen?
1	Red	Convertible	USA	Yes
2	Red	Convertible	USA	No
3	Red	Convertible	USA	Yes
4	Yellow	Convertible	USA	No
5	Yellow	Convertible	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	USA	No
9	Red	Convertible	Imported	No
10	Red	Convertible	Imported	Yes

will a
red SUV made in the U.S.
get stolen?

$$\Pr(\text{Stolen=yes} \mid \text{color=Red, type=SUV, made\_in=USA}) = \boxed{\frac{5}{10} \times \frac{\mathbf{0}}{5} \times \frac{2}{5} \times \frac{1}{5} = \mathbf{0}}$$

$$= \Pr(\text{Stolen}) \times \Pr(\text{Red} \mid \text{Stolen}) \times \Pr(\text{USA} \mid \text{Stolen}) \times \Pr(\text{SUV} \mid \text{Stolen})$$

$$\Pr(\text{Stolen=Yes}) = \frac{5}{10}$$

$$\Pr(\text{Made\_In = USA} \mid \text{Stolen=Yes}) = \frac{2}{5}$$

$$\boxed{\Pr(\text{Color=Red} \mid \text{Stolen=Yes}) = \frac{\mathbf{0}}{5}}$$

$$\Pr(\text{Type = SUV} \mid \text{Stolen=Yes}) = \frac{1}{5}$$

What would happen if there were no Red examples in the training set?

Would always estimate zero probability!

Naive Bayes Classifier - Problem #1

Instance #	Color	Type	Made_In	Class/Label: Likely to be stolen?
1	Red	Convertible	USA	Yes
2	Red	Convertible	USA	No
3	Red	Convertible	USA	Yes
4	Yellow	Convertible	USA	No
5	Yellow	Convertible	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	USA	No
9	Red	Convertible	Imported	No
10	Red	Convertible	Imported	Yes

will a
red SUV made in the U.S.
get stolen?

$$\Pr(\text{Stolen=yes} \mid \text{color=Red, type=SUV, made\_in=USA}) = \frac{5}{10} \times \frac{\mathbf{0}}{5} \times \frac{2}{5} \times \frac{1}{5} = \mathbf{0}$$

$$= \Pr(\text{Stolen}) \times \Pr(\text{Red} \mid \text{Stolen}) \times \Pr(\text{USA} \mid \text{Stolen}) \times \Pr(\text{SUV} \mid \text{Stolen})$$

$$\Pr(\text{Stolen=Yes}) = \frac{5}{10}$$

$$\Pr(\text{Made\_In = USA} \mid \text{Stolen=Yes}) = \frac{2}{5}$$

$$\Pr(\text{Color=Red} \mid \text{Stolen=Yes}) = \frac{\mathbf{0}}{5}$$

$$\Pr(\text{Type = SUV} \mid \text{Stolen=Yes}) = \frac{1}{5}$$

Possible solution: Laplace Smoothing

→ **adds (to numerator) 1**

→ **adds (to denominator) the number of possible values of the attribute**

e.g., |Color| = |{Red, Yellow}| = 2

$$\Pr(\text{Color=Red} \mid \text{Stolen=Yes}) = \frac{\#\text{stolenRedCars} + 1}{\#\text{stolenCars} + 2} = \frac{0 + 1}{5 + 2} = \frac{1}{7}$$

Naive Bayes Classifier - Problem #1

Instance #	*Color*	*Type*	*Made_In*	Class/Label: Likely to be stolen?
1	Red	Convertible	USA	**Yes**
2	Red	Convertible	USA	**No**
3	Red	Convertible	USA	**Yes**
4	Yellow	Convertible	USA	**No**
5	Yellow	Convertible	Imported	**Yes**
6	Yellow	SUV	Imported	**No**
7	Yellow	SUV	Imported	**Yes**
8	Yellow	SUV	USA	**No**
9	Red	Convertible	Imported	**No**
10	Red	Convertible	Imported	**Yes**

will a
red SUV made in the U.S.
get stolen?

$$\Pr(\text{Stolen=yes} \mid \text{color=Red, type=SUV, made\_in=USA}) = \frac{5}{10} \times \frac{\mathbf{0}}{5} \times \frac{2}{5} \times \frac{1}{5} = \mathbf{0}$$

$$= \Pr(\text{Stolen}) \times \Pr(\text{Red} \mid \text{Stolen}) \times \Pr(\text{USA} \mid \text{Stolen}) \times \Pr(\text{SUV} \mid \text{Stolen})$$

$$\Pr(\text{Stolen=Yes}) = \frac{5}{10} \qquad \Pr(\text{Made\_In = USA} \mid \text{Stolen=Yes}) = \frac{2}{5}$$

$$\Pr(\text{Color=Red} \mid \text{Stolen=Yes}) = \cancel{\frac{0}{5}} \quad \frac{0+1}{5+2} = \frac{1}{7} \qquad \Pr(\text{Type = SUV} \mid \text{Stolen=Yes}) = \frac{1}{5}$$

Possible solution: Laplace Smoothing

- → **adds (to numerator) 1**
- → **adds (to denominator) the number of possible values of the attribute**

e.g., |Color| = |{Red, Yellow}| = 2

$$\Pr(\text{Color=Red} \mid \text{Stolen=Yes}) = \frac{\#\text{stolenRedCars} + 1}{\#\text{stolenCars} + 2} = \frac{0+1}{5+2} = \frac{1}{7}$$

Naive Bayes Classifier - Problem #1

Instance #	*Color*	*Type*	*Made_In*	**Class/Label: Likely to be stolen?**
1	Red	Convertible	USA	**Yes**
2	Red	Convertible	USA	**No**
3	Red	Convertible	USA	**Yes**
4	Yellow	Convertible	USA	**No**
5	Yellow	Convertible	Imported	**Yes**
6	Yellow	SUV	Imported	**No**
7	Yellow	SUV	Imported	**Yes**
8	Yellow	SUV	USA	**No**
9	Red	Convertible	Imported	**No**
10	Red	Convertible	Imported	**Yes**

will a
red SUV made in the U.S.
get stolen?

$$\Pr(\text{Stolen=yes} \mid \text{color=Red, type=SUV, made\_in=USA}) = \boxed{\frac{5}{10} \times \frac{\mathbf{0}}{5} \times \frac{2}{5} \times \frac{1}{5} = \mathbf{0}}$$

$$= \Pr(\text{Stolen}) \times \Pr(\text{Red} \mid \text{Stolen}) \times \Pr(\text{USA} \mid \text{Stolen}) \times \Pr(\text{SUV} \mid \text{Stolen})$$

$$\Pr(\text{Stolen=Yes}) = \cancel{\frac{5}{10}} \quad \frac{5+1}{10+2} = \frac{6}{12}$$

$$\Pr(\text{Made\_In = USA} \mid \text{Stolen=Yes}) = \cancel{\frac{2}{5}} \quad \frac{2+1}{5+2} = \frac{3}{7}$$

$$\boxed{\Pr(\text{Color=Red} \mid \text{Stolen=Yes}) = \cancel{\frac{0}{5}}} \quad \frac{0+1}{5+2} = \frac{1}{7}$$

$$\Pr(\text{Type = SUV} \mid \text{Stolen=Yes}) = \cancel{\frac{1}{5}} \quad \frac{1+1}{5+2} = \frac{2}{7}$$

Possible solution: Laplace Smoothing

→ **adds (to numerator) 1**

→ **adds (to denominator) the number of possible values of the attribute**

e.g., |Color| = |{Red, Yellow}| = 2

$$\Pr(\text{Color=Red} \mid \text{Stolen=Yes}) = \frac{\#\text{stolenRedCars} + 1}{\#\text{stolenCars} + 2} = \frac{0+1}{5+2} = \frac{1}{7}$$

Naive Bayes Classifier - Problem #2

Instance #	*Color*	*Type*	*Made_In*	Class/Label: Likely to be stolen?
1	Red	Convertible	USA	Yes
2	Red	Convertible	USA	No
3	Red	Convertible	USA	Yes
4	Yellow	Convertible	USA	No
5	Yellow	Convertible	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	USA	No
9	Red	Convertible	Imported	No
10	Red	Convertible	Imported	Yes

will a
red SUV made in the U.S.
get stolen?

$$\Pr(\text{Stolen=yes} \mid \text{color=Red, type=SUV, made\_in=USA}) = \frac{5}{10} \times \frac{3}{5} \times \frac{2}{5} \times \frac{1}{5} = \frac{30}{10 \times 5^3}$$

$$= \Pr(\text{Stolen}) \times \Pr(\text{Red} \mid \text{Stolen}) \times \Pr(\text{USA} \mid \text{Stolen}) \times \Pr(\text{SUV} \mid \text{Stolen})$$

$$\Pr(\text{Stolen=no} \mid \text{color=Red, type=SUV, made\_in=USA}) = \frac{5}{10} \times \frac{2}{5} \times \frac{3}{5} \times \frac{2}{5} = \frac{60}{10 \times 5^3}$$

$$= \Pr(\text{Not\_Stolen}) \times \Pr(\text{Red} \mid \text{Not\_Stolen}) \times \Pr(\text{USA} \mid \text{Not\_Stolen}) \times \Pr(\text{SUV} \mid \text{Not\_Stolen})$$

which is larger?

Multiplying many small numbers between 0 and 1 (fractions)

Rapidly gets close to zero

Problems losing precision due to floating point representation

E.g.: $$\left(\frac{1}{1000} \times \frac{35}{550} \times \frac{52}{550} \times \frac{27}{550} \times \frac{67}{550} \times \frac{181}{550}\right) = .00000000956$$

Naive Bayes Classifier - Problem #2

Instance #	Color	Type	Made_In	Class/Label: Likely to be stolen?
1	Red	Convertible	USA	Yes
2	Red	Convertible	USA	No
3	Red	Convertible	USA	Yes
4	Yellow	Convertible	USA	No
5	Yellow	Convertible	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	USA	No
9	Red	Convertible	Imported	No
10	Red	Convertible	Imported	Yes

will a
red SUV made in the U.S.
get stolen?

$$\Pr(\text{Stolen=yes} \mid \text{color=Red, type=SUV, made\_in=USA}) = \log\left(\frac{5}{10} \times \frac{3}{5} \times \frac{2}{5} \times \frac{1}{5}\right)$$

$$= \Pr(\text{Stolen}) \times \Pr(\text{Red} \mid \text{Stolen}) \times \Pr(\text{USA} \mid \text{Stolen}) \times \Pr(\text{SUV} \mid \text{Stolen})$$

$$\Pr(\text{Stolen=no} \mid \text{color=Red, type=SUV, made\_in=USA}) = \log\left(\frac{5}{10} \times \frac{2}{5} \times \frac{3}{5} \times \frac{2}{5}\right)$$

$$= \Pr(\text{Not\_Stolen}) \times \Pr(\text{Red} \mid \text{Not\_Stolen}) \times \Pr(\text{USA} \mid \text{Not\_Stolen}) \times \Pr(\text{SUV} \mid \text{Not\_Stolen})$$

which is larger?

Possible solution: to compare the logarithm of those quantities!

E.g.: $\log\left(\frac{1}{1000} \times \frac{35}{550} \times \frac{52}{550} \times \frac{27}{550} \times \frac{67}{550} \times \frac{181}{550}\right) = ?$

Naive Bayes Classifier - Problem #2

Instance #	*Color*	*Type*	*Made_In*	Class/Label: Likely to be stolen?
1	Red	Convertible	USA	Yes
2	Red	Convertible	USA	No
3	Red	Convertible	USA	Yes
4	Yellow	Convertible	USA	No
5	Yellow	Convertible	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	USA	No
9	Red	Convertible	Imported	No
10	Red	Convertible	Imported	Yes

will a
red SUV made in the U.S.
get stolen?

$$\Pr(\text{Stolen=yes} \mid \text{color=Red, type=SUV, made\_in=USA}) = \log\left(\frac{5}{10} \times \frac{3}{5} \times \frac{2}{5} \times \frac{1}{5}\right)$$

$$= \Pr(\text{Stolen}) \times \Pr(\text{Red} \mid \text{Stolen}) \times \Pr(\text{USA} \mid \text{Stolen}) \times \Pr(\text{SUV} \mid \text{Stolen})$$

$$\Pr(\text{Stolen=no} \mid \text{color=Red, type=SUV, made\_in=USA}) = \log\left(\frac{5}{10} \times \frac{2}{5} \times \frac{3}{5} \times \frac{2}{5}\right)$$

$$= \Pr(\text{Not\_Stolen}) \times \Pr(\text{Red} \mid \text{Not\_Stolen}) \times \Pr(\text{USA} \mid \text{Not\_Stolen}) \times \Pr(\text{SUV} \mid \text{Not\_Stolen})$$

which is larger?

E.g.: $\log\left(\frac{1}{1000} \times \frac{35}{550} \times \frac{52}{550} \times \frac{27}{550} \times \frac{67}{550} \times \frac{181}{550}\right)$

$$= \log\left(\frac{1}{1000}\right) + \log\left(\frac{35}{550}\right) + \log\left(\frac{52}{550}\right) + \log\left(\frac{27}{550}\right) + \log\left(\frac{67}{550}\right) + \log\left(\frac{181}{550}\right)$$

Naive Bayes Classifier - Problem #2

Instance #	*Color*	*Type*	*Made_In*	Class/Label: Likely to be stolen?
1	Red	Convertible	USA	Yes
2	Red	Convertible	USA	No
3	Red	Convertible	USA	Yes
4	Yellow	Convertible	USA	No
5	Yellow	Convertible	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	USA	No
9	Red	Convertible	Imported	No
10	Red	Convertible	Imported	Yes

will a
red SUV made in the U.S.
get stolen?

$$\Pr(\text{Stolen=yes} \mid \text{color=Red, type=SUV, made\_in=USA}) = \boxed{\log\left(\frac{5}{10} \times \frac{3}{5} \times \frac{2}{5} \times \frac{1}{5}\right)}$$

$$= \Pr(\text{Stolen}) \times \Pr(\text{Red} \mid \text{Stolen}) \times \Pr(\text{USA} \mid \text{Stolen}) \times \Pr(\text{SUV} \mid \text{Stolen})$$

$$\Pr(\text{Stolen=no} \mid \text{color=Red, type=SUV, made\_in=USA}) = \boxed{\log\left(\frac{5}{10} \times \frac{2}{5} \times \frac{3}{5} \times \frac{2}{5}\right)}$$

$$= \Pr(\text{Not\_Stolen}) \times \Pr(\text{Red} \mid \text{Not\_Stolen}) \times \Pr(\text{USA} \mid \text{Not\_Stolen}) \times \Pr(\text{SUV} \mid \text{Not\_Stolen})$$

which is larger?

$$= \log\left(\frac{1}{1000}\right) + \log\left(\frac{35}{550}\right) + \log\left(\frac{52}{550}\right) + \log\left(\frac{27}{550}\right) + \log\left(\frac{67}{550}\right) + \log\left(\frac{181}{550}\right)$$

~~Multiplying~~ Adding many small numbers between 0 and 1 (fractions)

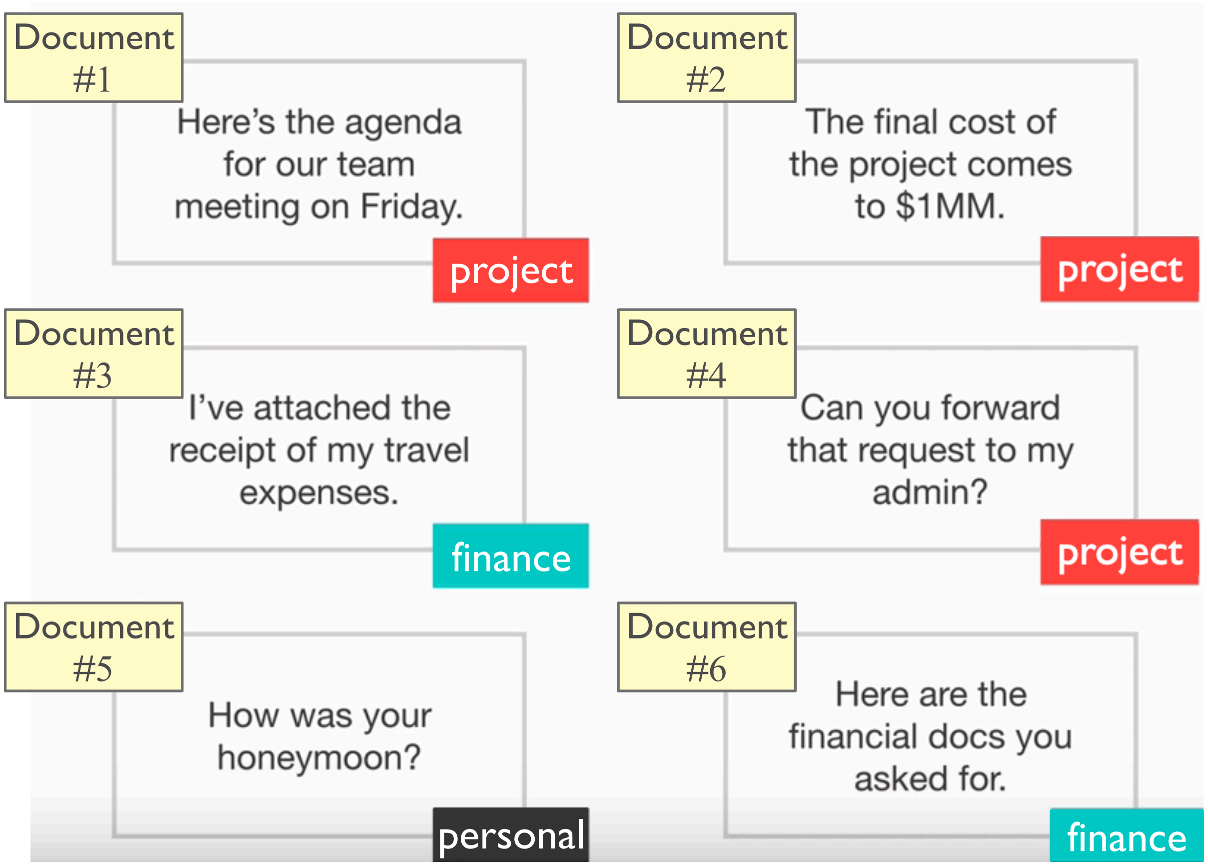
Fewer problems with precision due to floating point operations

Classifying (text) documents using Naive Bayes

Document Classification

Goal

Analyze contents of a document/article/paper/newspaper and identify/predict its class (e.g., "project", "finance", "personal")



Document Classification

- Formally, each document is represented by a vector of attributes
  - each attribute is associated with one possible word that may appear
  - the length of the vector is equal to the length of the **vocabulary**

- Consider the following vocabulary:

$$\mathbf{V} = \{\text{password, expired, send, review, conference, account, paper}\}$$

$$|\mathbf{V}| = 7 \text{ words}$$

How to represent a new document using a vector of this type?

Representing Documents (Naive Bayes Classification)

- Consider the following vocabulary:

$$\mathbf{V} = \{\text{password, expired, send, review, conference, account, paper}\}$$

- Assume a new document containing the following content:

"password expired: review account password"

- Bernoulli model: a document is represented by a binary vector
  each entry indicates if a given word appears or not in the document

b = (1,1,0,1,0,1,0)

	password	expired	send	review	conference	account	paper
doc1	1	1	0	1	0	1	0

- Multinomial model: a document is represented by a vector with integer components
  each entry indicates the frequency of a word in the document

m = (2,1,0,1,0,1,0)

	password	expired	send	review	conference	account	paper
doc1	**2**	1	0	1	0	1	0

Document Classification:

Bernoulli Naive Bayes

Document Classification - Bernoulli Naive Bayes

Consider the following document, b, represented with the Bernoulli model:

b

	password	expired	send	review	conference	account	paper
doc1	1	1	0	1	0	1	0

What is the probability of observing exactly those N=7 words in a document?

$$\Pr(\mathbf{b}) = \prod_{i=1}^{7} b_i \times \Pr(w_i) + (1 - b_i) \times (1 - \Pr(w_i))$$

Document Classification - Bernoulli Naive Bayes

Consider the following document, b, represented with the Bernoulli model:

b

	password	expired	send	review	conference	account	paper
doc1	**1**	**1**	**0**	**1**	**0**	**1**	**0**

What is the probability of observing exactly those N=7 words in a document?

$$\Pr(\mathbf{b}) = \prod_{i=1}^{7} b_i \times \Pr(w_i) + (1 - b_i) \times (1 - \Pr(w_i))$$

The i^{th} element of document **b**

$= \mathbf{1}$ if the i^{th} word in the vocabulary appears in **b**

$= \mathbf{0}$ otherwise

Document Classification - Bernoulli Naive Bayes

Consider the following document, b, represented with the Bernoulli model:

b

	password	expired	send	review	conference	account	paper
doc1	**1**	**1**	**0**	**1**	**0**	**1**	**0**

What is the probability of observing exactly those N=7 words in a document?

$$\Pr(\mathbf{b}) = \prod_{i=1}^{7} b_i \times \Pr(w_i) + (1 - b_i) \times (1 - \Pr(w_i))$$

Probability that the word w_i appears on a document

Document Classification - Bernoulli Naive Bayes

Consider the following document, b, represented with the Bernoulli model:

b

	password	expired	send	review	conference	account	paper
doc1	**1**	**1**	**0**	**1**	**0**	**1**	**0**

What is the probability of observing exactly those N=7 words in a document?

$$\Pr(\mathbf{b}) = \prod_{i=1}^{7} b_i \times \Pr(w_i) + (1 - b_i) \times (1 - \Pr(w_i))$$

$$= \Pr(w_1 = \text{password}) \times \Pr(w_2 = \text{expired}) \times (1 - \Pr(w_3 = \text{send})) \times \ldots$$

Document Classification - Bernoulli Naive Bayes

Intuition...

- Consider tossing a coin N=7 times
- Assume the probability of tossing a head (H) is Pr(H) = 70%
  the probability of tossing a tail (T) is Pr(T) = (1 - Pr(H)) = 30%

$$\mathbf{e} = (H, T, T, H, H, H, T)$$

What is the probability of observing exactly those N=7 tosses in **e**?

$$\Pr(\mathbf{e}) = \Pr(H)\,\Pr(T)\,\Pr(T)\,\Pr(H)\,\Pr(H)\,\Pr(H)\,\Pr(T)$$
$$= \Pr(H)\,(1 - \Pr(H))\,(1 - \Pr(H))\,\Pr(H)\,\Pr(H)\,\Pr(H)\,(1 - \Pr(H))$$

Let's re-write **e** as a binary vector:

entry e_i = 1 if the toss landed heads
entry e_i = 0 if the toss landed tails

$$\mathbf{e} = (1, 0, 0, 1, 1, 1, 0)$$

$$\Pr(\mathbf{e}) = \prod_{i=1}^{7} e_i \times \Pr(H) + (1 - e_i) \times \Pr(T) = \prod_{i=1}^{7} e_i \times \Pr(H) + (1 - e_i) \times (1 - \Pr(H))$$

Document Classification - Bernoulli Naive Bayes

Consider the following document, b, represented with the Bernoulli model:

b

	password	expired	send	review	conference	account	paper
doc1	1	1	0	1	0	1	0

What is the probability of observing exactly those N=7 words in a document?

$$\Pr(\mathbf{b}) = \prod_{i=1}^{7} b_i \times \Pr(w_i) + (1 - b_i) \times (1 - \Pr(w_i))$$

What is the probability of the document belonging to class y_i ?

$$\Pr(y_i \mid \mathbf{Doc}) = \Pr(y_i \mid \mathbf{b}) = \frac{\Pr(y_i)\ \Pr(\mathbf{b} \mid y_i)}{\cancel{\Pr(\mathbf{b})}}$$

Document Classification - Bernoulli Naive Bayes

Consider the following document, b, represented with the Bernoulli model:

b

	password	expired	send	review	conference	account	paper
doc1	1	1	0	1	0	1	0

What is the probability of observing exactly those N=7 words in a document?

$$\Pr(\mathbf{b}) = \prod_{i=1}^{7} b_i \times \Pr(w_i) + (1 - b_i) \times (1 - \Pr(w_i))$$

What is the probability of the document belonging to class y_i ?

$$\Pr(y_i \mid \mathbf{Doc}) \propto \Pr(y_i \mid \mathbf{b}) = \Pr(y_i)\ \Pr(\mathbf{b} \mid y_i)$$

$$= \Pr(y_i) \prod_{k=1}^{7} b_k \times \Pr(w_k \mid y_i) + (1 - b_k) \times (1 - \Pr(w_k \mid y_i))$$

Document Classification - Bernoulli Naive Bayes

Consider the following document, b, represented with the Bernoulli model:

b

	password	expired	send	review	conference	account	paper
doc1	**1**	**1**	**0**	**1**	**0**	**1**	**0**

What is the probability of the document belongs to class y_i ?

$$\Pr(y_i \mid \mathbf{b}) = \Pr(y_i) \prod_{k=1}^{7} b_k \times \Pr(w_k \mid y_i) + (1 - b_k) \times (1 - \Pr(w_k \mid y_i))$$

More generally:

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr\left(w_k \mid y_i\right) + (1 - b_k)(1 - \Pr\left(w_k \mid y_i\right)) \right]$$

- $|V|$ is the size of the vocabulary (i.e., the size of **b**)
- b_k is an entry in **b** representing whether the k^{th} word shows up in the document (0 or 1)
- $\Pr(w_k \mid y_i)$ is the probability of the k^{th} word appearing in documents of the class y_i
- $1 - \Pr(w_k \mid y_i)$ is the probability of the k^{th} word **not** appearing in documents of the class y_i

Document Classification - Bernoulli Naive Bayes

Consider the following document, b, represented with the Bernoulli model:

b

	password	expired	send	review	conference	account	paper
doc1	**1**	**1**	**0**	**1**	**0**	**1**	**0**

What is the probability of the document belongs to class y_i ?

$$\Pr(y_i \mid \mathbf{b}) = \Pr(y_i) \prod_{k=1}^{7} b_k \times \Pr(w_k \mid y_i) + (1 - b_k) \times (1 - \Pr(w_k \mid y_i))$$

More generally:

$$\Pr(y_i \mid Doc) = \Pr(y_i) \prod^{|V|} \left[b_k \Pr(w_k \mid y_i) + (1 - b_k)(1 - \Pr(w_k \mid y_i)) \right]$$

How to estimate these probabilities based on training data?

- $\Pr(w_k \mid y_i)$ is the probability of the k^{th} word appearing in documents of the class y_i
- $1 - \Pr(w_k \mid y_i)$ is the probability of the k^{th} word **not** appearing in documents of the class y_i

Document Classification - Bernoulli Naive Bayes

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr\left(w_k \mid y_i\right) + (1 - b_k)\left(1 - \Pr\left(w_k \mid y_i\right)\right) \right]$$

- V is the vocabulary of all known words
- N is the total number of documents available
- $N(y_i)$ is the number of documents belonging to class y_i
- $n(w_k, y_i)$ is the number of documents belonging to class y_i where the word w_k appears
- $\Pr(w_k \mid y_i)$ is the probability of the k^{th} word appearing in documents of the class y_i
- $1 - \Pr(w_k \mid y_i)$ is the probability of the k^{th} word **not** appearing in documents of the class y_i

Document Classification - Bernoulli Naive Bayes

$$\Pr(y_i \mid \text{Doc}) = \boxed{\Pr(y_i)} \prod_{k=1}^{|V|} \left[b_k \Pr\left(w_k \mid y_i\right) + (1 - b_k)\left(1 - \Pr\left(w_k \mid y_i\right)\right) \right]$$

- V is the vocabulary of all known words
- N is the total number of documents available
- $N(y_i)$ is the number of documents belonging to class y_i
- $n(w_k, y_i)$ is the number of documents belonging to class y_i where the word w_k appears
- $\Pr(w_k \mid y_i)$ is the probability of the k^{th} word appearing in documents of the class y_i
- $1 - \Pr(w_k \mid y_i)$ is the probability of the k^{th} word **not** appearing in documents of the class y_i

$$\boxed{\Pr(y_i)} = \frac{N(y_i)}{N}$$

percentage of
documents
of class y_i

Document Classification - Bernoulli Naive Bayes

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \boxed{\Pr\left(w_k \mid y_i\right)} + (1 - b_k)\left(1 - \boxed{\Pr\left(w_k \mid y_i\right)}\right) \right]$$

- V is the vocabulary of all known words
- N is the total number of documents available
- $N(y_i)$ is the number of documents belonging to class y_i
- $n(w_k, y_i)$ is the number of documents belonging to class y_i where the word w_k appears
- $\Pr(w_k \mid y_i)$ is the probability of the k^{th} word appearing in documents of the class y_i
- $1 - \Pr(w_k \mid y_i)$ is the probability of the k^{th} word **not** appearing in documents of the class y_i

$$\Pr(y_i) = \frac{N(y_i)}{N}$$

percentage of documents of class y_i

$$\boxed{\Pr\left(w_k \mid y_i\right)} = \frac{n(w_k, y_i)}{N(y_i)}$$

percentage of documents of class y_i that contain the word w_k

Bernoulli Naive Bayes Classification

- Consider the problem of classifying a document
  - it may belong to the "Sports" class (**SP**), or to the "Informatics" (**INF**) class…
  - depending on the topics being discussed in the document

- We analyzed the training set (composed of many documents) and identified the following vocabulary

$$V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$$

$$|V| = 8$$

Example adapted from the lecture notes of Hiroshi Shimodaira (University of Edinburgh)

Bernoulli Naive Bayes Classification

Attributes of a document (words that may appear in it) → $V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$

$$\mathbf{B}_{SP} = \begin{array}{c} \begin{array}{cccccccc} w_1 & w_2 & w_3 & w_4 & w_5 & w_6 & w_7 & w_8 \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \end{pmatrix} \end{array}$$

Number of documents belonging to the class **SP** ("Sports")

Number of words in the vocabulary (considering documents of all classes)

Bernoulli Naive Bayes Classification

Attributes of a document (words that may appear in it) $\rightarrow$ $V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$

$$\mathbf{B}_{SP} = \begin{array}{c} \begin{array}{cccccccc} w_1 & w_2 & w_3 & w_4 & w_5 & w_6 & w_7 & w_8 \end{array} \\ \left.\begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \end{pmatrix}\right\} \text{6 documents} \end{array}$$

$$\mathbf{B}_{INF} = \left.\begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}\right\} \text{5 documents}$$

Bernoulli Naive Bayes Classification

Attributes of a document (words that may appear in it) $\rightarrow$ $V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$

$$\mathbf{B}_{SP} = \begin{array}{c} \begin{array}{cccccccc} w_1 & w_2 & w_3 & w_4 & w_5 & w_6 & w_7 & w_8 \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \end{pmatrix} \end{array} \Bigg\} \text{ 6 documents}$$

How often the word w_3 appears in all documents

$$\mathbf{B}_{INF} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix} \Bigg\} \text{ 5 documents}$$

Bernoulli Naive Bayes Classification

Attributes of a document (words that may appear in it) → $V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$

$$\begin{array}{cc} & \begin{array}{cccccccc} w_1 & w_2 & w_3 & w_4 & w_5 & w_6 & w_7 & w_8 \end{array} \\ \mathbf{B}_{SP} = & \left.\begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \end{pmatrix}\right\} \text{6 documents} \end{array}$$

Description of the 3[rd] document belonging to the class "Sports"

$$\mathbf{B}_{INF} = \left.\begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}\right\} \text{5 documents}$$

Bernoulli Naive Bayes Classification

Attributes of a document (words that may appear in it) → $V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$

$$\mathbf{B}_{SP} = \left.\begin{array}{c} \begin{array}{cccccccc} w_1 & w_2 & w_3 & w_4 & w_5 & w_6 & w_7 & w_8 \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \end{pmatrix} \end{array}\right\} \textbf{6 documents}$$

$$\mathbf{B}_{INF} = \left.\begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}\right\} \textbf{5 documents}$$

Description of the 2nd document belonging to the class "Informatics"

Bernoulli Naive Bayes Classification

Estimating the probabilities needed to compute $\Pr(y_i \mid \text{Doc})$

$$\Pr(y_i \mid \text{Doc}) = \underline{\Pr(y_i)} \prod_{k=1}^{|V|} \left[b_k \Pr\left(w_k \mid y_i\right) + (1 - b_k)(1 - \Pr\left(w_k \mid y_i\right)) \right]$$

$$V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$$

$$\Pr(y_i) = \frac{N(y_i)}{N}$$

percentage of documents of class y_i

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

6 documents

$$\mathbf{B}_{\text{SP}} = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \end{pmatrix}$$

5 documents

$$\mathbf{B}_{\text{INF}} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Bernoulli Naive Bayes Classification

Estimating the probabilities needed to compute $\Pr(y_i \mid \text{Doc})$

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \underline{\Pr(w_k \mid y_i)} + (1 - b_k)(1 - \underline{\Pr(w_k \mid y_i)}) \right]$$

$$V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$$

$$\Pr(w_k \mid y_i) = \frac{n(w_k, y_i)}{N(y_i)}$$

- $N(y_i)$ is the number of documents belonging to class y_i
- $n(w_k, y_i)$ is the number of documents belonging to class y_i where the word w_k appears
- $\Pr(w_k \mid y_i)$ is the probability of the k^{th} word appearing in documents of the class y_i

6 documents

$$\mathbf{B}_{\text{SP}} = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \end{pmatrix}$$

5 documents

$$\mathbf{B}_{\text{INF}} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Bernoulli Naive Bayes Classification

Estimating the probabilities needed to compute $\Pr(y_i \mid \text{Doc})$

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \underline{\Pr(w_k \mid y_i)} + (1 - b_k)(1 - \underline{\Pr(w_k \mid y_i)}) \right]$$

$$V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$$

$$\Pr(w_k \mid y_i) = \frac{n(w_k, y_i)}{N(y_i)}$$

- $N(y_i)$ is the number of documents belonging to class y_i
- $n(w_k, y_i)$ is the number of documents belonging to class y_i where the word w_k appears
- $\Pr(w_k \mid y_i)$ is the probability of the k^{th} word appearing in documents of the class y_i

	$n(\cdot, \text{SP})$
w_1	3
w_2	1
w_3	2
w_4	3
w_5	3
w_6	4
w_7	4
w_8	4

$n(\cdot, \text{INF})$
1
3
3
1
1
1
3
1

6 documents

$$\mathbf{B}_{\text{SP}} = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \end{pmatrix}$$

5 documents

$$\mathbf{B}_{\text{INF}} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Bernoulli Naive Bayes Classification

Estimating the probabilities needed to compute $\Pr(y_i \mid \text{Doc})$

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \underline{\Pr(w_k \mid y_i)} + (1 - b_k)(1 - \underline{\Pr(w_k \mid y_i)}) \right]$$

$$V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$$

$$\Pr(w_k \mid y_i) = \frac{n(w_k, y_i)}{N(y_i)}$$

- $N(y_i)$ is the number of documents belonging to class y_i
- $n(w_k, y_i)$ is the number of documents belonging to class y_i where the word w_k appears
- $\Pr(w_k \mid y_i)$ is the probability of the k^{th} word appearing in documents of the class y_i

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

6 documents

$$\mathbf{B}_{\text{SP}} = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \end{pmatrix}$$

5 documents

$$\mathbf{B}_{\text{INF}} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr\left(w_k \mid y_i\right) + (1 - b_k)(1 - \Pr\left(w_k \mid y_i\right)) \right]$$

Consider a new document described by $\mathbf{b}$ = (0, 1, 1, 0, 1, 0, 1, 0). To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

Bernoulli Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k \mid y_i) + (1 - b_k)(1 - \Pr(w_k \mid y_i)) \right]$$

Consider a new document described by **b** = (0, 1, 1, 0, 1, 0, 1, 0). To which class does it belong?

$\Pr(y_i = \text{SP}) = 6/11$

$\Pr(y_i = \text{INF}) = 5/11$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\Pr(y = \text{SP} \mid \mathbf{b}) = 6/11 \times$$
$$(1\text{-}0)\ (1 - 3/6) \times$$
$$1 \quad 1/6 \times$$
$$1 \quad 2/6 \times$$
$$(1\text{-}0)\ (1 - 3/6) \times$$
$$1 \quad 3/6 \times$$
$$(1\text{-}0)\ (1 - 4/6) \times$$
$$1 \quad 4/6 \times$$
$$(1\text{-}0)\ (1 - 4/6)$$
$$= 0.00028$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k \mid y_i) + (1 - b_k)(1 - \Pr(w_k \mid y_i)) \right]$$

Consider a new document described by **b** = (0, 1, 1, 0, 1, 0, 1, 0). To which class does it belong?

$\Pr(y_i = \text{SP}) = 6/11$

$\Pr(y_i = \text{INF}) = 5/11$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\Pr(y = \text{SP} \mid \mathbf{b}) = 6/11 \times$$
$$(1\text{-}0)\,(1 - 3/6) \times$$
$$1 \quad 1/6 \times$$
$$1 \quad 2/6 \times$$
$$(1\text{-}0)\,(1 - 3/6) \times$$
$$1 \quad 3/6 \times$$
$$(1\text{-}0)\,(1 - 4/6) \times$$
$$1 \quad 4/6 \times$$
$$(1\text{-}0)\,(1 - 4/6)$$
$$= 0.00028$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr\left(w_k \mid y_i\right) + (1 - b_k)(1 - \Pr\left(w_k \mid y_i\right)) \right]$$

Consider a new document described by **b** = (0, 1, 1, 0, 1, 0, 1, 0). To which class does it belong?

$\Pr(y_i = \text{SP}) = 6/11$

$\Pr(y_i = \text{INF}) = 5/11$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\begin{aligned}
\Pr(y = \text{SP} \mid \mathbf{b}) = \quad & 6/11 \times \\
& (1\text{-}0)\ (1 - 3/6) \times \\
& 1 \quad 1/6 \times \\
& 1 \quad 2/6 \times \\
& (1\text{-}0)\ (1 - 3/6) \times \\
& 1 \quad 3/6 \times \\
& (1\text{-}0)\ (1 - 4/6) \times \\
& 1 \quad 4/6 \times \\
& (1\text{-}0)\ (1 - 4/6) \\
& = 0.00028
\end{aligned}$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr\left(w_k \mid y_i\right) + (1 - b_k)\left(1 - \Pr\left(w_k \mid y_i\right)\right) \right]$$

Consider a new document described by $\mathbf{b}$ = (0, 1, 1, 0, 1, 0, 1, 0). To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\begin{aligned}
\Pr(y = \text{SP} \mid \mathbf{b}) = \quad & 6/11 \times \\
& (1\text{-}0)\ (1 - 3/6) \times \\
& 1 \quad 1/6 \times \\
& 1 \quad 2/6 \times \\
& (1\text{-}0)\ (1 - 3/6) \times \\
& 1 \quad 3/6 \times \\
& (1\text{-}0)\ (1 - 4/6) \times \\
& 1 \quad 4/6 \times \\
& (1\text{-}0)\ (1 - 4/6) \\
& = 0.00028
\end{aligned}$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr\left(w_k \mid y_i\right) + (1 - b_k)(1 - \Pr\left(w_k \mid y_i\right)) \right]$$

Consider a new document described by **b** = (0, 1, 1, 0, 1, 0, 1, 0). To which class does it belong?

$\Pr(y_i = \text{SP}) = 6/11$

$\Pr(y_i = \text{INF}) = 5/11$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\begin{aligned}
\Pr(y = \text{SP} \mid \mathbf{b}) = \quad & 6/11 \times \\
& (1\text{-}0)\ (1 - 3/6) \times \\
& 1 \quad 1/6 \times \\
& 1 \quad 2/6 \times \\
& (1\text{-}0)\ (1 - 3/6) \times \\
& 1 \quad 3/6 \times \\
& (1\text{-}0)\ (1 - 4/6) \times \\
& 1 \quad 4/6 \times \\
& (1\text{-}0)\ (1 - 4/6) \\
& = 0.00028
\end{aligned}$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k \mid y_i) + (1 - b_k)(1 - \Pr(w_k \mid y_i)) \right]$$

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$\Pr(y_i = \text{SP}) = 6/11$

$\Pr(y_i = \text{INF}) = 5/11$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\begin{aligned}
\Pr(y = \text{SP} \mid \mathbf{b}) = \; & 6/11 \times \\
& (1\text{-}0)\ (1 - 3/6) \times \\
& 1 \quad 1/6 \times \\
& 1 \quad 2/6 \times \\
& (1\text{-}0)\ (1 - 3/6) \times \\
& 1 \quad 3/6 \times \\
& (1\text{-}0)\ (1 - 4/6) \times \\
& 1 \quad 4/6 \times \\
& (1\text{-}0)\ (1 - 4/6) \\
& = 0.00028
\end{aligned}$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k \mid y_i) + (1 - b_k)(1 - \Pr(w_k \mid y_i)) \right]$$

Consider a new document described by **b** = (0, 1, 1, 0, 1, 0, 1, 0). To which class does it belong?

$\Pr(y_i = \text{SP}) = 6/11$

$\Pr(y_i = \text{INF}) = 5/11$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\begin{aligned}
\Pr(y = \text{SP} \mid \mathbf{b}) = \quad & 6/11 \times \\
& (1\text{-}0)\ (1 - 3/6) \times \\
& 1 \quad 1/6 \times \\
& 1 \quad 2/6 \times \\
& (1\text{-}0)\ (1 - 3/6) \times \\
& 1 \quad 3/6 \times \\
& (1\text{-}0)\ (1 - 4/6) \times \\
& 1 \quad 4/6 \times \\
& (1\text{-}0)\ (1 - 4/6) \\
& = 0.00028
\end{aligned}$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr\left(w_k \mid y_i\right) + (1 - b_k)(1 - \Pr\left(w_k \mid y_i\right)) \right]$$

Consider a new document described by **b** = (0, 1, 1, 0, 1, 0, 1, 0). To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\begin{aligned}
\Pr(y = \text{SP} \mid \mathbf{b}) = \quad & 6/11 \times \\
& (1\text{-}0)\ (1 - 3/6) \times \\
& 1 \quad 1/6 \times \\
& 1 \quad 2/6 \times \\
& (1\text{-}0)\ (1 - 3/6) \times \\
& 1 \quad 3/6 \times \\
& (1\text{-}0)\ (1 - 4/6) \times \\
& 1 \quad 4/6 \times \\
& (1\text{-}0)\ (1 - 4/6) \\
& = 0.00028
\end{aligned}$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k \mid y_i) + (1 - b_k)(1 - \Pr(w_k \mid y_i)) \right]$$

Consider a new document described by **b** = (0, 1, 1, 0, 1, 0, 1, 0). To which class does it belong?

$\Pr(y_i = \text{SP}) = 6/11$

$\Pr(y_i = \text{INF}) = 5/11$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\begin{aligned}
\Pr(y = \text{SP} \mid \mathbf{b}) = \quad & 6/11 \times \\
& (1\text{-}0)\ (1 - 3/6) \times \\
& 1 \quad 1/6 \times \\
& 1 \quad 2/6 \times \\
& (1\text{-}0)\ (1 - 3/6) \times \\
& 1 \quad 3/6 \times \\
& (1\text{-}0)\ (1 - 4/6) \times \\
& 1 \quad 4/6 \times \\
& (1\text{-}0)\ (1 - 4/6) \\
& = 0.00028
\end{aligned}$$

 Bruno Castro da Silva

Bernoulli Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr\left(w_k \mid y_i\right) + (1 - b_k)(1 - \Pr\left(w_k \mid y_i\right)) \right]$$

Consider a new document described by **b** = (0, 1, 1, 0, 1, 0, 1, 0). To which class does it belong?

$\Pr(y_i = \text{SP}) = 6/11$

$\Pr(y_i = \text{INF}) = 5/11$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\begin{aligned}
\Pr(y = \text{SP} \mid \mathbf{b}) = \quad & 6/11 \times \\
& (1\text{-}0)\ (1 - 3/6) \times \\
& 1 \quad 1/6 \times \\
& 1 \quad 2/6 \times \\
& (1\text{-}0)\ (1 - 3/6) \times \\
& 1 \quad 3/6 \times \\
& (1\text{-}0)\ (1 - 4/6) \times \\
& 1 \quad 4/6 \times \\
& (1\text{-}0)\ (1 - 4/6) \\
& \boxed{= 0.00028}
\end{aligned}$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k \mid y_i) + (1 - b_k)(1 - \Pr(w_k \mid y_i)) \right]$$

Consider a new document described by **b** = (0, 1, 1, 0, 1, 0, 1, 0). To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\Pr(y = \text{SP} \mid \mathbf{b}) = 0.00028$$

$$\Pr(y = \text{INF} \mid \mathbf{b}) = 5/11 \times$$

(1-0) (1 - 1/5) ×
1 3/5 ×
1 3/5 ×
(1-0) (1 - 1/5) ×
1 1/5 ×
(1-0) (1 - 1/5) ×
1 3/5 ×
(1-0) (1 - 1/5)
= 0.008

Bernoulli Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr\left(w_k \mid y_i\right) + (1 - b_k)\left(1 - \Pr\left(w_k \mid y_i\right)\right) \right]$$

Consider a new document described by **b** = (0, 1, 1, 0, 1, 0, 1, 0). To which class does it belong?

$\Pr(y_i = \text{SP}) = 6/11$

$\Pr(y_i = \text{INF}) = 5/11$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$\Pr(y = \text{SP} \mid \mathbf{b}) = 0.00028$

$$\begin{aligned}\Pr(y = \text{INF} \mid \mathbf{b}) = \quad & 5/11 \times \\ & (1\text{-}0)\,(1 - 1/5) \times \\ & 1 \quad 3/5 \times \\ & 1 \quad 3/5 \times \\ & (1\text{-}0)\,(1 - 1/5) \times \\ & 1 \quad 1/5 \times \\ & (1\text{-}0)\,(1 - 1/5) \times \\ & 1 \quad 3/5 \times \\ & (1\text{-}0)\,(1 - 1/5) \\ & = 0.008\end{aligned}$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k \mid y_i) + (1 - b_k)(1 - \Pr(w_k \mid y_i)) \right]$$

Consider a new document described by **b** = (0, 1, 1, 0, 1, 0, 1, 0). To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
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& (1\text{-}0)\ (1 - 1/5) \quad \times \\
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The new document seems to belong to the "Informatics" class

Machine Learning
CMPSCI 589

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Naive Bayes (2/2)
Bernoulli Naive Bayes

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